

Equation of a line

Key Points

The vector equation of a line is

$$r = a + \lambda d$$

Where r is the coordinate, a is a (starting) point on the line and d is the direction vector of the line. λ is the amount travelled in the direction of the line

To convert to cartesian rearrange for λ which is a constant

$$\frac{x - a_1}{d_1} = \frac{y - a_2}{d_2} = \frac{z - a_3}{d_3}$$

Note. One may be constant

Basic Skills

Q1. Given that $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

- a) Find the coordinate when $\lambda = 2$
- b) Find the coordinate when $\lambda = -2$
- c) Write down $x(\lambda), y(\lambda)$
- d) Find λ such that $x = 20$ and hence find the corresponding y

Q2. Given that $A=(3,4)$ and $B=(5,-1)$ form a line

- a) Find direction vector along the line
- b) Write the vector equation of the line
- c) Explain why the direction vector is like the gradient

Competency Skills

Q4. A line goes through the coordinates $A=(-1,2,5)$ and $B=(6,4,-3)$

- a) Find direction vector along the line and hence write a vector equation of the line
- b) Write a different (but equivalent) vector equation of the line
- c) Write functions for the coordinates in terms of your parameter and hence state the cartesian equation

Q5. Write the equation $\frac{x-5}{2} = \frac{y-2}{-3} = \frac{z+1}{5}$ in vector form

Q6. Write the cartesian equation of the line going through the origin in the direction $2i - j + 4k$

Q7. Convert the line $x = 2, y = 2z$ into vector form

Q8. Find if the line $r = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \lambda \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$ goes through the point $A=(10,-3,-1)$

Advanced Skills

Q9. a) Find in vector form line l_1 which goes through the point $(3,-2,1)$ and $(5,0,2)$

b) Convert to vector form the line l_2 with cartesian equation $3x - 6 = 2y + 1 = z - 5$

c) By using the directions of the two lines explain why the lines are not parallel

d) By using the scalar product determine the angle between the direction of both lines

e) Determine if, and if so where, the lines intersect

Q10. The relative position vector of a plane, to the control tower t seconds after take-off is given by $r = 3ti + t^2j + (e^t - 2)k$ find the distance the plane travels in the 10 seconds after take-off

Extension

Q11. For two lines in 3D space do they always intersect when they are not travelling in a parallel

Hint: Consider the number of dimensions compared to the parameters controlling the position of a line